LESSON 7.5

Represent Rational Numbers on the Coordinate Plane

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can plot and identify points on the coordinate plane using ordered pairs.

WHY THIS MATTERS Captain Vega left a treasure map, but instead of an 'X marks the spot,' she hid the directions as ordered pairs! To find the treasure, you need to plot each point on the coordinate plane and connect them in order. The first clue says: Start at (2, 1), then go to (5, 4), then (8, 1).

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Coordinate plane *Spanish: Plano cartesiano*	A grid with a line going across and a line going up to plot points.	x-axis across, y-axis up
Ordered pair *Spanish: Par ordenado*	Two numbers (x, y) that tell where a point is on a grid.	(−3, 4) is an ordered pair
Origin *Spanish: Origen*	The point (0, 0) where the two grid lines cross.	every plot starts here
Quadrant *Spanish: Cuadrante*	One of the four parts of a coordinate grid, numbered I to IV.	Quadrant II is upper left
Reflection *Spanish: Reflexión*	A flipped, mirror image across a line.	(3, 2) reflected over the x-axis becomes (3, -2) — same x, opposite y
Integer *Spanish: Número entero*	Whole numbers and their opposites, like -2, -1, 0, 1, 2.	..., -3, -2, -1, 0, 1, 2, 3, ...

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Plot (−5, −4). Name its quadrant.

1. Start at the origin, (0, 0).
2. Move 5 units left, because x = −5.
3. Move 4 units down, because y = −4.

Answer: (−, −) is Quadrant III

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now let's plot (2, 1). Where do we start? At the origin (0, 0).

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The ordered pair _____ tells me to move _____ and then _____ .
- The point _____ lands in Quadrant _____ because its signs are _____ .
- I always start at the origin because _____ .
- A point sitting on the _____ -axis has a coordinate of _____ .

Word bank coordinate plane | ordered pair | origin | quadrant | x-axis | y-axis | x-coordinate | y-coordinate | horizontal | vertical | left | right

WATCH OUT!

A common mistake

A common mistake in Graph on the Coordinate Plane is swapping the order of the coordinates

LESSON 7.5

Represent Rational Numbers on the Coordinate Plane

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can plot and identify points on the coordinate plane using ordered pairs.

WHY THIS MATTERS Captain Vega left a treasure map, but instead of an 'X marks the spot,' she hid the directions as ordered pairs! To find the treasure, you need to plot each point on the coordinate plane and connect them in order. The first clue says: Start at (2, 1), then go to (5, 4), then (8, 1).

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

A point is at (6, 2). What does the 6 tell you?

(A) Move 6 units to the right

(B) Move 6 units up

(C) Move 6 units to the left

(D) The point is in Quadrant 6

HINT In (6, 2), the 6 is the FIRST number. First and second positions have different jobs.

SHOW YOUR WORK

2 **Write the name of the correct group on each line.**

Put these steps in order for plotting the point (4, 5).

Groups:

_____ Start at the origin (0, 0)

_____ Move 4 units to the right along the x-axis

_____ Move 5 units up from that spot

_____ Place a dot at the point (4, 5)

HINT Every plotting routine begins at the same home spot before any moving happens.

3 **Write the letter of the matching item on each line.**

Match each ordered pair to its description.

_____ (0, 0)	(A) Origin
_____ (4, 2)	(B) Right 4, Up 2
_____ (1, 6)	(C) Right 1, Up 6
_____ (5, 5)	(D) Right 5, Up 5
_____ (3, 0)	(E) On the x-axis
_____ (0, 4)	(F) On the y-axis

HINT Each pair is (x, y): first number = right, second = up. A 0 means you don't move in that direction.

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

Which ordered pair names a point 3 units right and 7 units up from the origin?

(A) (0, 7)
(B) (3, 7)
(C) (3, 3)
(D) (7, 3)

HINT Re-read the moves: 3 units right, then 7 units up. Which move comes first in an ordered pair?

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

What are the coordinates of the origin?

(A) (0, 0)
(B) (0, 1)
(C) (1, 1)
(D) (1, 0)

HINT The origin is the exact spot where the x-axis and y-axis cross. Picture that crossing point.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Clue** Plot the point (5, 2).
2. **Start** Start at the origin (0, 0). Good.
3. **Move up** Move 5 units UP along the y-axis.
4. **Move right** Move 2 units to the right. Place the dot at (2, 5).

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Which direction does the first number in (x, y) tell you to move — right or up?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

You plotted a triangle. How did the ordered pairs tell you exactly where to place each point?

SENTENCE STARTER The first number in the ordered pair tells me ___ and the second number tells me ___. The point (5, 4) means ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 7.5

Represent Rational Numbers on the Coordinate Plane

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can plot and identify points on the coordinate plane using ordered pairs.

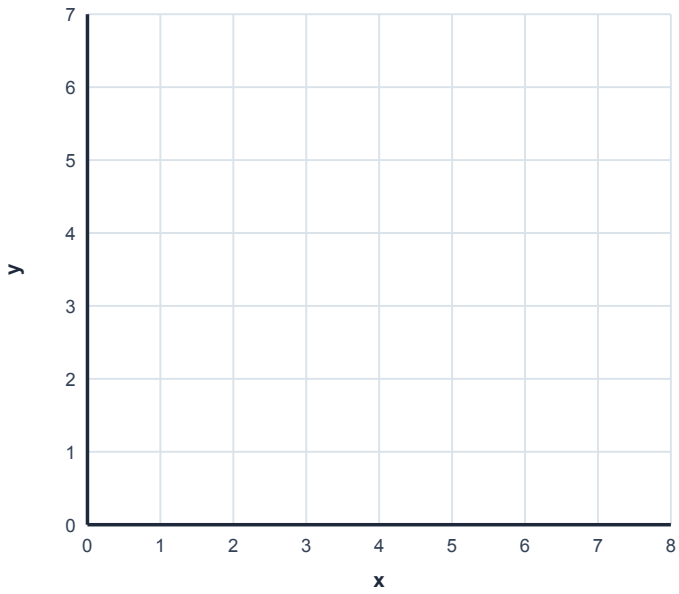
WHY THIS MATTERS Captain Vega left a treasure map, but instead of an 'X marks the spot,' she hid the directions as ordered pairs! To find the treasure, you need to plot each point on the coordinate plane and connect them in order. The first clue says: Start at (2, 1), then go to (5, 4), then (8, 1).

PART 1 Understand the idea

models and meaning

1 **Plot and label each point on the grid.**

Plot these points to reveal a hidden shape: (1, 1), (1, 5), (6, 5), (6, 1)



SHOW YOUR THINKING

2 **Write the letter of the matching item on each line.**

Match each point to its quadrant or axis.

_____ (3, 2)
_____ (-4, 5)
_____ (-2, -6)
_____ (7, -1)
_____ (0, 3)
_____ (5, 0)

(A) Quadrant I
(B) Quadrant II
(C) Quadrant III
(D) Quadrant IV
(E) y-axis
(F) x-axis

3 **Complete the table. Show how you found each value.**

Complete the table. Write the ordered pair for each description.

Description	Ordered Pair
Right 7, Up 2	
Right 0, Up 5	
Right 4, Up 0	
Right 3, Up 3	

SHOW YOUR WORK

4 **Write the name of the correct group on each line.**

Sort these ordered pairs by their x-coordinate from smallest to largest.

Groups:

_____ (5, 3)
_____ (1, 7)
_____ (3, 2)
_____ (0, 4)
_____ (7, 1)

PART 2 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Clue** Plot the point (4, 0).
2. **Move right** Move 4 units to the right along the x-axis. Good.
3. **Move up** Move 4 units up, because the pair has a 4 in it.
4. **Plot the point** Place the dot at (4, 4).

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

You plotted a triangle. How did the ordered pairs tell you exactly where to place each point?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 7.5

Represent Rational Numbers on the Coordinate Plane

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can plot and identify points on the coordinate plane using ordered pairs.

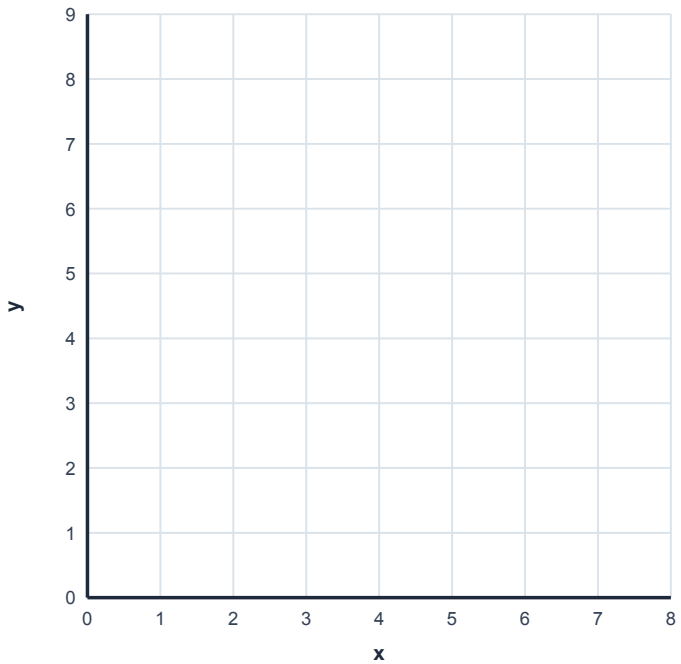
WHY THIS MATTERS Captain Vega left a treasure map, but instead of an 'X marks the spot,' she hid the directions as ordered pairs! To find the treasure, you need to plot each point on the coordinate plane and connect them in order. The first clue says: Start at (2, 1), then go to (5, 4), then (8, 1).

PART 1 Understand the idea

models and meaning

1 **Plot and label each point on the grid.**

Plot the points (2, 3), (2, 7), (6, 7), (6, 3). Connect them. Name the shape and find its perimeter using the coordinates.



SHOW YOUR THINKING

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Miscounted Grid Lines

1. **Ordered pair** (7, 3)
2. **Move along the x-axis** Start at the origin (0, 0) and count 7 units to the right.
3. **Move along the y-axis** Count up from the x-axis: 1, 2, 3, 4 — stop on 4.
4. **Plot the point** Place the dot at (7, 4).

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A student says that (3, 5) and (5, 3) are the same point because they use the same numbers. Is the student correct? Explain why or why not, and describe where each point is located on the coordinate plane.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

You plotted a triangle. How did the ordered pairs tell you exactly where to place each point?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help